# Dragon Kingdom

The Care and Training of Your Wall Dragon - for a young keeper and their grown-ups

## Item 1: Do not plug in your dragon before reading this

Congratulations! You are now the keeper of a genuine, pedigreed **Wall Dragon**. Unlike wild dragons, which take up three bedrooms and set the curtains on fire, a domesticated Wall Dragon needs only a suitable wall screw fitted into the keyhole-shaped mounting slot on its back, and a steady 5 V snack through the USB-C power port at the bottom.

Your new dragon is a sensitive creature and performs a strict waking ritual whenever power is connected. **Do not rush a waking dragon:**

1. Ask a grown-up to hang your dragon on its wall screw and connect the power supply provided or specified by your maker.
2. Watch its snout: first, it blinks **one dim blue eye five times** to make sure the wall has not moved.
3. Next, it lights a few white bars along the bottom. **It is testing the strength of the cord** to choose a suitable brightness limit. After holding those bars for two seconds of quiet pride, your dragon wakes in full colour.

## Know your dragon (79 breeds in one)

Your dragon didn't hatch out of just any boulder! Because keeping seventy-nine full-sized dragons in one bedroom is frowned upon, your Wall Dragon is trained in **Chameleon Shifting**. With a single pat, it can channel **79 famous dragons** from the tribes-plus up to **eight rare hybrids** invented by you.

**Keep your setup card.** Every dragon uses the same QR card. You may write its Wi-Fi name in the optional blank. The setup password is **dragon123** (all lowercase).

# Simple obedience

Section One - basic manners and bedroom etiquette

## Training your dragon

Nobody likes a surly, misbehaving dragon. Amazingly, a Wall Dragon responds to training without a leash, muzzle, or fire extinguisher.

- **STAY:** Hang your dragon on its wall screw, look it squarely in the snout, and say, **STAY**. Incredibly, your dragon will obey on the very first try. Peek back into the room an hour later and you will find it is still on the wall. Praise your dragon.
- **COME:** Stand across the room and call your dragon's name. Assuming it is properly hung on its screw, it will not move. Do not give up; walk slowly toward the wall. Astonishingly, as you walk toward your dragon, you will notice it coming closer!
- **SPEAK & FETCH:** Don't be ridiculous. Wall Dragons have no ears, no feet, and a deep philosophical objection to barking. They speak entirely in light.

## Your three touch commands

Pat the touch button on the right side of your dragon to give it commands-no phone or internet required:

- **One pat (Next!):** Shifts into the next dragon in your collection, or wakes a sleeping dragon.
- **Two quick pats (Play Dead / Sleep):** Sends your dragon to sleep in a dark cave, or wakes it back up.
- **Long hold (Purr-Brighten / Dim):** Hold your finger on the pad to adjust its glow in small steps. Let go and hold again to go the other way.

## Nocturnal manners

Your dragon is not afraid of the dark. When you turn off the bedroom lights, it gently lowers its glow to keep quiet watch while you sleep. It adjusts again as the room brightens. Your chosen Glow setting and the available power still apply; use Sleep when you want the lights off.

# Connect a phone

Section Two - telepathy for grown-up keepers

## Open Dragon Kingdom

While your dragon is happy being patted on its right-side touch button without knowing what the internet is, grown-ups can speak to it through a phone browser. No app or coding is needed.

1. Read the **Setup & manual** QR guide while you have internet. Power on one new dragon and wait for its waking ritual.
2. Open phone Wi-Fi settings and choose **Dragon-Kingdom-** followed by six letters/numbers. If several appear, power off the other dragons during setup.
3. Enter **dragon123** (all lowercase).
4. If the phone warns that this network has no internet, choose **Stay connected**. (Dragons do not read emails.)
5. The setup page may appear automatically. Otherwise scan **Open dragon** on the card or rear label. Fallback: **http://192.168.4.1**.
6. Play without logging in. Open **Parent settings** and unlock with **dragon123** for Wi-Fi, backups, updates and reset.

You can play and edit right away over the dragon's own network. Home Wi-Fi is optional. Updated firmware uses **dragon123** even if an older card shows a random password. Your home Wi-Fi keeps its own private password.

## Optional: teach it your home Wi-Fi

1. Unlock **Parent settings** with **dragon123**, then open **Home Wi-Fi**.
2. Tap **Find my Wi-Fi** and choose your **2.4 GHz** home network (or type its name).
3. Enter your home Wi-Fi password. Use **Show password** to check your spelling.
4. Tap **Connect my dragon** and wait for the result.
5. Open **Connection & power** above the play controls and bookmark its **.local** address.
6. Setup Wi-Fi closes about 15 seconds after success. Switch your phone back to home Wi-Fi, open that address in your normal browser, and bookmark it. A numeric fallback is also shown.

# The dawn stretch

Section Three - teaching your dragon rooster duty

## Set your wake-up time

Roosters are noisy and drop feathers on the floor. A trained Wall Dragon wakes you silently by painting the wall behind its wings with sunrise light.

1. Open **Sunrise alarm** on the phone page and wait for its saved schedule to load.
2. Tick **Wake me every day**.
3. Choose your **Wake-up time** and **Sunrise length (minutes)**.
4. Tap **Save alarm** and check the confirmation message.

Your dragon begins warming its entire wings-from the lowest wing scales to the wing tips-from deep ember-orange to morning gold before wake-up time. For example, **7:00** with a **20-minute** sunrise starts at **6:40**.

It works even while the dragon is asleep. At wake-up time, the wings hold their morning glow for **30 minutes**, then return to their previous state. Available power still caps brightness, and the dragon never makes a sound.

## Rehearse or dismiss the sunrise

- **Try a 30-second sunrise** previews the dawn stretch, then returns to normal after a brief hold.
- **Stop sunrise** (or any pat / play button) tells your dragon "I'm awake!" without cancelling tomorrow's alarm.
- To turn off rooster duty completely, untick **Wake me every day** and tap **Save alarm**.

## Dragon timekeeping

Dragons do not wear wristwatches. Saving an alarm borrows the clock and daylight-saving rules from your phone. If unplugged, the dragon remembers your schedule, but after a power cut it needs home Wi-Fi internet time or this page opened again (**Set clock from my phone**) before it knows what hour it is. Open and save the alarm once a year or after changing time zones.

# Roosts and collections

Section Four - The Great Sanctuary and personal teams

## All or a personal roost

Every dragon lives in **All** (The Great Sanctuary). It holds all 79 pedigreed dragons plus each custom hybrid currently saved on your device. By ancient dragon law, **All** cannot be renamed or deleted, and you can use it forever without making a profile.

If you want smaller hand-picked teams for different moods, use **My profile** to switch between up to **three named profiles** (personal roosts).

When a named profile is active, **My dragon** and single pats on the touch sensor cycle only through that profile's favourites and its own creations. The first dragon in the list greets you after startup.

## Build a personal roost

1. Open **My favourites & starting glow**.
2. Enter a **New profile name** and tap **Add profile**.
3. Tick up to **12** favourite dragons in the order you want them to appear. (Untick and tick again to move one to the end.)
4. Pick a **Starting glow**, then tap **Save profile**.

Your own creations appear right after your ticked favourites. A brand-new empty profile will say **Add favourites or make a dragon**, and play buttons rest until you put at least one dragon in that roost. Profiles and creations survive unplugging; temporary play tricks do not overwrite your saved startup glow.

## Closing a roost

To remove a named profile, select it, open its settings, and tap **Delete this profile**. Never worry about losing your hybrids: any custom dragons hatched in that roost fly safely back into **All**.

# Hatching a hybrid

Section Five - custom colours, body parts and elemental tricks

## Breed and save your own dragon

1. Tap **Make a dragon** (or **Edit or copy custom dragon**) and choose a starter in **Start from**.
2. Give your creation a **Dragon name**.
3. Tap a body region on the dragon picture, or pick one from **Dragon part**. The picture is a colour guide; save to see the result on your wall.
4. Choose its **Colour** and **Magic on this part** (such as Glitter, Shimmer, Breathe, or Rainbow).
5. Paint as many parts as you like, then tap **Save my dragon**.

You can colour the **face, body, tail, horns, wing edges, and wing interiors** separately. Untouched parts keep their natural markings, and **Use original colour for this part** restores a region's base scales.

The editor shows your dragon's shape so you can choose parts naturally. Tap **Save my dragon** to watch your creation come alive on the wall!

## Edit, copy, or retire a hybrid

- Select your dragon in **My dragon** and tap **Edit or copy custom dragon** to groom or rename it.
- **Save my dragon** updates a hybrid you own; **Save a new copy** hatches a twin.
- Editing one of the 79 pedigreed dragons (or a hybrid from another profile) always saves a new copy so originals stay untouched.
- Your wall holds up to **eight custom dragons** in total. When all eight perches are full, edit or delete an old creation to make room.

## Amusing elemental tricks

On the play screen, tap **Glitter**, **Fire**, **Breathe**, or **Rainbow**. Your dragon will perform the trick tirelessly until you tap **Back to my dragon**. Use **Glow** to adjust brightness, or **Sleep / Wake my dragon** for nap time. Wait a beat for each trick to land before tapping the next button.

# Dragon health care

Section Six - moving house and first aid for common ailments

## Moving house (The power-cycle setup)

If you change Wi-Fi routers or want a direct phone connection again, teach your dragon the **power-cycle setup**:

1. Unplug the power cord from the bottom of the dragon.
2. Reconnect power normally; setup Wi-Fi opens at startup. No touch hold is needed.
3. Wait for startup, then join the **Dragon-Kingdom-XXXXXX** Wi-Fi network with **dragon123** and open **http://192.168.4.1**.

This keeps your saved home Wi-Fi and creations intact. To erase home Wi-Fi completely, unlock Parent settings, open **Open setup Wi-Fi again**, tap **Forget home Wi-Fi**, then power off and on.

## When your dragon seems confused

- **Phone says "no internet":** Completely normal on the dragon's private network. Stay connected and open **http://192.168.4.1**.
- **The .local bookmark won't open:** Make sure your phone is on the same home Wi-Fi as the dragon, or try its numeric address. Guest networks often forbid pets from talking to phones-use a regular home network or the power-cycle setup.
- **Dragon Wi-Fi never appears:** If the power brick is weak, the dragon saves its energy for touch lighting and keeps Wi-Fi off. Ask a grown-up to check the power supply specified by the maker.
- **Page says "Please wait for your dragon":** An action is still being processed. Wait for it to finish before trying another control.
- **Dragon is unexpectedly dark or blinking red:** Check **Sleep** and **Glow** first (remember it dims in dark rooms). If it shows a dim red code at startup or keeps restarting, unplug it and contact the maker.
- **Interrupted saves or lost time:** A failed save keeps your draft open in the editor. If the page asks to reload after a firmware update or collection change, note your unsaved colour choices first. If the alarm waits after a power cut, tap **Set clock from my phone**.

# Keeper's certificate

Section Seven - care rules and official registration

## Feeding and care (For grown-ups)

The bottom USB-C port is power-only. Always feed your dragon through that port using the power supply provided or specified by your maker. Keep your dragon indoors and dry, handle its cord gently, and leave enclosure service to a grown-up following the maker’s recovery instructions, and never attach unapproved power sources.

## A good dragon keeper remembers

Carved around your dragon's frame, except along the bottom, is the motto **Darkness will rise to bring the light**-every hard climb or quiet night comes right before you shine.

- **Touch play needs no network**-even without a phone, Wi-Fi, or internet.
- **Keep your setup card** and bookmark your dragon's **.local** address.
- **All** is the protected Great Sanctuary; named profiles are personal roosts.
- **Save your work** in the editor before unplugging the dragon.
- **Polite brightness:** Your dragon automatically adjusts its glow for dark bedrooms and the strength of its power supply, so 100% means "as bright as current conditions safely allow."
- Never throw your dragon across a pond, and never turn it loose in the wild. Remember: if you take care of your Wall Dragon, your Wall Dragon will chase away the dark for you.

## Official pedigree record

First dragon's name: _____

Keeper's name: _____

Phone bookmark (.local): _____

Maker / Dragon healer contact: _____

Dragon Wi-Fi and Parent settings password: **dragon123**. Play needs no login. Keep your home Wi-Fi password private.

**About this edition:** Dragon Kingdom shipping user manual, version 1.5, 25 October 2026. For firmware v0.3.0 and beta 3 or later. Keeper edition for the completed enclosure: right-side touch button, rear keyhole mounting slot and bottom USB-C power port.

# Grown-up service notes

USB recovery and refitting the backplate

## Open for recovery

The bottom USB-C port is power-only. Never connect laptop USB and external/main power together. Wi-Fi updates need no opening.

1. Disconnect the bottom product USB-C power cable before connecting laptop USB.
2. Remove the screw at the top rear of the box.
3. Using a screwdriver through the rear keyhole, push the release tab until the backplate pops open.
4. Connect the laptop USB-C cable to the ESP32 on the main board in the centre of the backplate.

## Easy Update: automatic USB updates

Choose **USB cable** in Easy Update on internet. Keep **USB Tools** with the app. Confirm USB-only power for all boards and enable **Auto update USB dragons**; signed files are prepared first. Connect one board, wait for the result, unplug and wait for **USB Auto ready** before the next. Older firmware updates; equal/newer versions are not rewritten. **Stop** finishes the current operation; errors stop Auto. Backups stay private. Unresponsive boards need maker-assisted recovery.

## Close after recovery

Disconnect laptop USB. Fit the backplate's bottom USB-C port and LDR (room-light sensor) into their box slots. Press the backplate top into place and replace the screw. Reconnect product power only after laptop USB is disconnected.

## Support details and power

**Support details** downloads firmware/board, uptime, startup, estimated power, calibration and connection/memory diagnostics for the maker. It excludes passwords, tokens, Wi-Fi network names and custom dragon names.

**Connection & power** shows home addresses, detected power mode and estimated startup limit. Computer USB is capped at **400 mA**. Product-supply mode means no computer USB was detected; its limit is an estimate, not a measured supply rating.

# Keep your dragon up to date

Easy Update - the recommended Windows/Mac method

For the shipped beta-3 HTTPS error and later updates. **Do not factory-reset.** Keep each dragon on its normal bottom USB-C supply; no laptop USB or opening the box.

**Download Easy Update:** https://sai021.github.io/dragon-kingdom/update/

Extract the ZIP. Open **Dragon Kingdom Easy Update.exe** (Windows) or **Dragon Kingdom Easy Update.app** (Mac). Keep the extracted files together.

## Open, wait, then update

For lamps **not on home Wi-Fi**. Offline mode is selected. Only use the batch button where all nearby dragons are yours.

1. Open the app on internet. It automatically downloads verified latest stable firmware. Wait until ready; opening installs nothing.
2. Choose **Find & update all dragons**. Allow OS Wi-Fi/location permission if asked. The app joins each setup network, checks identity and updates older firmware. Current/newer lamps are skipped.
3. Wait for the results report. Parent password **dragon123** is filled in and editable. Each update saves a private backup and checks startup/settings; transfer needs no internet.

## Already on home Wi-Fi?

Choose **Home Wi-Fi**: put computer and lamp on the same network, then **Check for updates** and **Find dragons**. Select yours (or **Add address**), then **Update selected dragon**.

Setup AP closes 15 seconds after joining home Wi-Fi; these lamps need Home Wi-Fi mode.

## Let the checks finish

Keep power and the app connected. It tries to reconnect and restore the original Wi-Fi after each lamp. Checks allow five minutes per lamp. **Stop** cancels safely; after finalization it finishes checking that lamp and skips the rest. If Wi-Fi restoration fails, the batch stops.

Offline startup uses the verified saved release, which may be older. Reopen on internet for fresh files. Check lighting/touch afterward; inspect the installed version before retrying failures. The guide includes manual and phone-only alternatives.